

Python Exercises for the Notebook

Write the Python code required to solve each problem in your NOTEBOOK.

Exercise 1: Basic Input and Type Casting

Write a program that asks the user for their current age in years. Calculate and print the approximate year they were born (assuming the current year is 2026). Ensure you properly cast the input string into an integer.

Exercise 2: Algorithmic Operators (Modulo)

Write a program using the modulo operator (%) to determine if a given number of days is a perfect number of weeks. Prompt the user for a number of days, and print the remaining leftover days after dividing into full 7-day weeks.

Exercise 3: Formatting with f-strings

Prompt the user for their first name, last name, and favorite programming language. Use a single f-string to print the following message: 'Hello, [First Name] [Last Name]! I see your favorite language is [Language].'

Exercise 4: F-Strings and Math

Ask the user for a product name, its price, and the quantity they wish to buy. Calculate the total cost. Use an f-string to print a receipt message that includes the product name and the total cost formatted to 2 decimal places (e.g., \$15.50).

Exercise 5: Sum of Digits

Read a natural number consisting of exactly two digits (e.g., 45) from the keyboard. Write a program that calculates and prints the sum of its digits. (Hint: use integer division // to extract the tens and modulo % to extract the units).

Exercise 6: Time Conversion

Read a natural number 'm' from the keyboard, representing a time duration in minutes. Write a program that converts this duration into hours and minutes, and prints the result using an f-string in the format: 'X hours and Y minutes'. (Hint: use integer division to find the number of hours and modulo for the remaining minutes).

Exercise 7: Ornaments

In a Christmas tree, there are 'a' white ornaments, twice as many red ornaments, and 3 fewer green ornaments than red ones. Write a program that reads the number 'a' (number of white ornaments) from the keyboard, then calculates and prints the total number of ornaments in the tree.